

Week 1 - Introduction to AI Engineering

Concepts:

Overview of AI Engineering, AI Engineering role and responsibilities

Project:

Create a personal AI Engineering portfolio

Outcome:

Understand the basics of AI Engineering and its applications

Week 2 - Mathematics for AI

Concepts:

Linear Algebra, Calculus, Probability and Statistics

Project:

Solve AI-related math problems

Outcome:

Understand the mathematical foundations of AI

Week 3 - Programming Fundamentals

Concepts:

Python programming, Data Structures and Algorithms

Project:

Build a simple AI-related project in Python

Outcome:

Develop programming skills for AI

Week 4 - Machine Learning Fundamentals

Concepts:

Supervised and Unsupervised Learning, Regression and Classification

Project:

Implement a simple machine learning model

Outcome:

Understand the basics of machine learning

Week 5 - Deep Learning Fundamentals

Concepts:

Neural Networks, Convolutional Neural Networks

Project:

Build a simple deep learning model

Outcome:

Understand the basics of deep learning

Week 6 - Natural Language Processing

Concepts:

Text Preprocessing, Sentiment Analysis

Project:

Implement a simple NLP task

Outcome:

Understand the basics of NLP

Week 7 - Computer Vision

Concepts:

Image Processing, Object Detection

Project:

Implement a simple computer vision task

Outcome:

Understand the basics of computer vision

Week 8 - Reinforcement Learning

Concepts:

Q-Learning, Policy Gradient Methods

Project:

Implement a simple reinforcement learning task

Outcome:

Understand the basics of reinforcement learning

Week 9 - AI Ethics and Bias

Concepts:

Fairness and Bias in AI, AI and Society

Project:

Analyze AI-related bias and fairness issues

Outcome:

Understand the importance of AI ethics and bias

Week 10 - AI Engineering Tools and Frameworks

Concepts:

TensorFlow, PyTorch

Project:

Build a project using a popular AI framework

Outcome:

Understand popular AI engineering tools and frameworks

Week 11 - AI Engineering Best Practices

Concepts:

Code Quality and Readability, Testing and Debugging

Project:

Implement best practices in an AI-related project

Outcome:

Understand AI engineering best practices

Week 12 - Capstone Project

Concepts:

Project:

Build a comprehensive AI-related project

Outcome:

Apply AI engineering skills to a real-world project
